

Thomas Pethick

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I am interested in understanding the structure underlying deep learning models and use this to build algorithms that learn efficiently and robustly. Two major parts of my work evolves around:

- Large-scale pre-training that exploits the structure of neural networks and is agnostic to scale, with a focus on time- and memory efficiency.
- Developing algorithms for fixed-point problems, which can be used to capture multi-player games, robust training, and recurrent architectures, where divergent and cycling dynamics are common.

I strongly believe that deep learning does not need to be an art, but that it can be made into a scientific discipline through both theoretical and empirical work.

Keywords: *Deep Learning, Optimization, Multi-agent Systems, Game-Theory, Robustness, Pre-training*

Education

École Polytechnique Fédérale de Lausanne (EPFL) **Lausanne (Switzerland)**
PhD in Electrical Engineering with Prof. Volkan Cevher (Partially funded by Google) *2019–2025*
EPFL Thesis Distinction (awarded to the 8% best PhD theses)

École Polytechnique Fédérale de Lausanne (EPFL) **Lausanne (Switzerland)**
MSc Exchange *2018–2019*
Incl. Information Theory & Coding, Information Theory & Signal Processing, Stochastic Calculus, Optimization for ML.

Technical University of Denmark (DTU) **Kongens Lyngby (Denmark)**
MSc Honours Programme in Computer Science and Engineering *2017–2019*
Thesis on "Scalable Gaussian Processes for Economic Models". Projects: "Multi-agent System for Reversible MultiBAN", "Epistemic Planning and Games", "Prover Programming" and "Environmental Sound Classification with SSL"

University of Illinois at Urbana-Champaign (UIUC) **Urbana-Champaign (United States)**
BSc Exchange *Fall 2016*
Incl. CUDA Programming, Machine Learning, System Programming

Technical University of Denmark (DTU) **Kongens Lyngby (Denmark)**
BSc in Computer Software Engineering *2014–2017*
Thesis on "MCMC and Probabilistic Programming Using Monads"

Research Internships

Amazon Research Internship **Tübingen (Germany)**
Multi-epoch pre-training of LLMs and memory-efficient optimization methods. *2023*

Swiss Federal Institute of Technology (ETH) **Zürich (Switzerland)**
"Ensembled Deep Network for Global Optimization" advised by Prof. Andreas Krause *Summer 2018*
Student Summer Research Fellowship Program, selected as one of 16 participants from among 1,000+ applicants.

Publications #alphabetical order, *first author contribution

Stable Optimization in Deep Learning: Geometry and Games (thesis)
Thomas Pethick*. 2025.

Training Neural Networks at Any Scale (under review)

Thomas Pethick*, Kimon Antonakopoulos, Antonio Silveti-Falls, Leena Chennuru Vankadara, and Volkan Cevher. 2025.

Generalized Gradient Norm Clipping & Non-Euclidean (L_0, L_1) -Smoothness **(Oral, 0.36% of all submissions)**
Thomas Pethick*, Wanyun Xie*, Mete Erdogan, Kimon Antonakopoulos, Tony Silveti-Falls, and Volkan Cevher
Neural Information Processing Systems (NeurIPS). 2025.

Training Deep Learning Models with Norm-Constrained LMOs **(Spotlight)**

Thomas Pethick*, Wanyun Xie, Kimon Antonakopoulos, Zhenyu Zhu, Antonio Silveti-Falls, and Volkan Cevher
International Conference on Machine Learning (ICML). 2025.

Efficient interpolation between extragradient and proximal methods for weak MVIs

Thomas Pethick*, Ioannis Mavrothalassitis, and Volkan Cevher

International Conference on Learning Representations (ICLR). 2025.

SAMPa: Sharpness-aware Minimization Parallelized

Wanyun Xie, **Thomas Pethick**, and Volkan Cevher

Neural Information Processing Systems (NeurIPS). 2024.

ν SAM: Memory-Efficient Sharpness-Aware Minimization via Nuclear Norm Constraints

Thomas Pethick*, Parameswaran Raman, Lenon Minorics, Mingyi Hong, Shoham Sabach, and Volkan Cevher

Transactions on Machine Learning Research (TMLR). 2025.

iFedDR: Auto-Tuning Local Computation with Inexact Douglas-Rachford Splitting in Federated Learning

Thomas Pethick*, Puya Latafat, Basile Lewandowski, Zheng Xu, Peter Kairouz, Panagiotis Patrinos, and Volkan Cevher. 2024.

Improving SAM Requires Rethinking its Optimization Formulation

Xie, Wanyun, Fabian Latorre, Kimon Antonakopoulos, **Thomas Pethick**, and Volkan Cevher

International Conference on Machine Learning (ICML). 2024.

Stable nonconvex-nonconcave training via linear interpolation (Spotlight)

Thomas Pethick*, Wanyun Xie, and Volkan Cevher

Advances in Neural Information Processing Systems (NeurIPS). 2023.

Mixed Nash for Robust Federated Learning

Xie, Wanyun, **Thomas Pethick**, Ali Ramezani-Kebrya, and Volkan Cevher

Transactions on Machine Learning Research (TMLR). 2023.

Solving stochastic weak Minty variational inequalities without increasing batch size

Thomas Pethick*, Olivier Fercoq, Puya Latafat, Panagiotis Patrinos, and Volkan Cevher

International Conference on Learning Representations (ICLR). 2023.

Revisiting adversarial training for the worst-performing class

Thomas Pethick*, Grigorios Chrysos, and Volkan Cevher

Transactions on Machine Learning Research (TMLR). 2023.

Finding Actual Descent Directions for Adversarial Training

Latorre, Fabian, Igor Krawczuk, Leello Tadesse Dadi, **Thomas Pethick**, and Volkan Cevher

International Conference on Learning Representations (ICLR). 2023.

Federated Learning under Covariate Shifts with Generalization Guarantees

Ramezani-Kebrya*, Ali, Fanghui Liu*, **Thomas Pethick***, Grigorios Chrysos, and Volkan Cevher

Transactions on Machine Learning Research (TMLR). 2023.

Escaping limit cycles: Global convergence for constrained nonconvex-nonconcave minimax problems (Spotlight)

Thomas Pethick*, Puya Latafat, Panagiotis Patrinos, Olivier Fercoq, and Volkan Cevher

International Conference on Learning Representations (ICLR). 2022.

Sifting through the noise: Universal first-order methods for stochastic variational inequalities

Antonakopoulos, Kimon, **Thomas Pethick**, Ali Kavis, Panayotis Mertikopoulos, and Volkan Cevher

Advances in Neural Information Processing Systems (NeurIPS). 2021.

Subquadratic overparameterization for shallow neural networks

Song, Chaehwan, Ali Ramezani-Kebrya, **Thomas Pethick**, Armin Eftekhari, and Volkan Cevher

Advances in Neural Information Processing Systems (NeurIPS). 2021.

A Process Calculus for Design and Modeling of Retro-Synthesis

#Gabriel Broholm, Maliina Hammeken, Marcus Skov Hansen, Alexander Juhl, Marc Storm Larsen, Jørgen Fischer Nilsson, **Thomas Pethick**, and Magnus Gether Sørensen

Information Modelling and Knowledge Bases XXX. 2019.

Supervised Students

Mete Erdogan (Intern): <i>Non-Euclidean Methods</i>	Fall 2025
Basile Lewandowski (Intern): <i>Proximal Machine Unlearning</i>	Spring 2024
Wanyun Xie (Master Thesis): <i>A Game-Theoretical Framework for Robust Federated Learning</i>	Spring 2022
Lorena Egger (Master Thesis): <i>Worst Performing Class on Neural Architecture Search</i>	Spring 2022
Abdülkadir Gökce (Semester Project & Intern): <i>Adversarial Dropout</i>	Fall 2021

Teaching Experience

Online Learning in Games (EPFL), <i>co-created and co-instructed</i>	Spring '23,'24
Mathematics of Data (EPFL), <i>teaching assistant</i>	Fall '20,'21,'22,'23
Reinforcement Learning (EPFL), <i>teaching assistant</i>	Spring 2022
Deep Learning (EPFL), <i>teaching assistant</i>	Spring 2021
Applied Machine Learning (EPFL), <i>teaching assistant</i>	Spring 2020
Online Convex Optimization reading group (EPFL), <i>organizer</i> , lecture notes on pethick.dk	Fall 2019
Introduction to Artificial Intelligence (DTU), <i>teaching assistant</i>	Spring 2018
Probability Theory (DTU), <i>teaching assistant</i>	Fall 2015

Academic Service

Conference reviewing: NeurIPS, ICML, ICLR	2022-present
Journal reviewing: TMLR	2024

Technical Knowledge

Languages: Python, JavaScript, F#, C/C++, Haskell, Java

ML libraries: PyTorch, JAX, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, CUDA

Technologies: Git, Slurm, Docker, REST APIs, SQL, NoSQL, LaTeX, Django, Flask, React, Node.js

Developer Experience

Software engineer 2010–2017

Prior to and during university I worked as a software developer across front-end, back-end and app development and have managed small teams in the process. Companies include Fødevarebanken, Rainmaking, and Birdback.

Open source software 2015–present

In my freetime I enjoy building open source incl. [Anki add-ons](#), [a cross-platform dropbox app for DTUs filesystem](#), [A scheduler for GPU jobs](#), and [a zoo of minimax problems](#). See [my github](#) for a complete list.

Awards

EPFL Thesis Distinction 2025

Awarded to the 8% best PhD theses.

Jørgen Klinth-Jensen award at Christianshavn's Gymnasium 2013

"This grant goes to a student who has shown great professional talent in many areas but at the same time generously shared his knowledge and his technical skill with others."